Student Course Registration System

Black-box and white-box testing report

Project overview

The project tests an in-memory course registration module against REQUIREMENTS.md. The supplied implementation is treated as the system under test. Expected outcomes remain requirement-based; no production-code fixes or test expectation changes were made.

Test execution summary

Run	Total	Passed	Failed
Initial black-box	58	43	15
Expanded black-box	59	43	16
White-box	20	18	2
Combined	79	61	18

All failures are retained because they expose behavior that conflicts with the SRS. Detailed transcripts and coverage evidence are saved under docs/evidence/.

1. Black-box test design

Equivalence Class Partitioning (ECP) divides inputs into classes expected to behave alike. Cases include valid and invalid current/course credits (FR-01, FR-02); empty, complete, partial, or absent prerequisites (FR-03); zero/positive/negative dues (FR-04, FR-15); duplicate status (FR-05); and eligible/ineligible availability paths (FR-06 through FR-09).

Boundary Value Analysis (BVA)

Boundary cases cover current credits 0 and 18, course credits 1 and 4, combined totals exactly 18 and just above, seats at capacity / one below / zero, and dues 0 versus 1. Input-validation boundaries also exercise invalid credits, capacity below 1, and enrollment outside 0..capacity (FR-02, FR-15).

Dimension	Representative input	SRS expected result
Credit total	14 current + 4 course	Allowed: total exactly 18
Credit total	15 current + 4 course	Rejected: total exceeds 18
Course boundary	1 and 4 credit hours	Both valid course-credit bounds
Seat boundary	enrolled = capacity	WAITLISTED
Seat boundary	enrolled < capacity	REGISTERED
Dues boundary	0 / 1 rupee	continue / DUES_BLOCK

Full test IDs, requirement traceability, conditions, and expected results are listed in docs/blackbox_test_design.xlsx.

2. Decision table and priority order

The system returns the first applicable outcome in the ordered checks below. A dash means a lower-priority condition does not affect that rule.

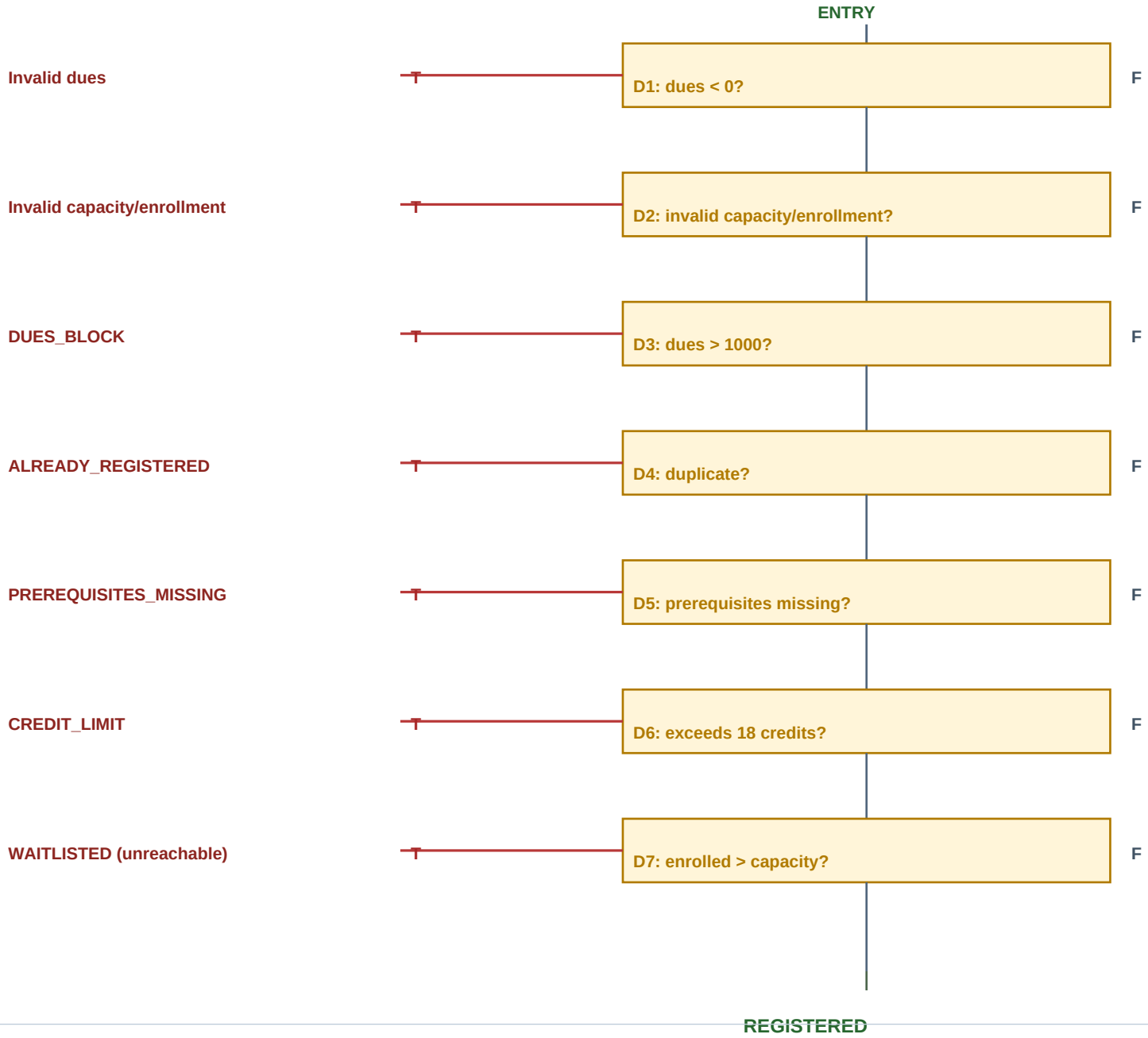
Rule	Dues	Duplicate	Prerequisites	Credit limit	Availability	Outcome
1	> 0	-	-	-	-	DUES_BLOCK
2	= 0	Yes	-	-	-	ALREADY_REGISTERED
3	= 0	No	Missing any	-	-	PREREQUISITES_MISSING
4	= 0	No	All met	Would exceed 18	-	CREDIT_LIMIT
5	= 0	No	All met	Fits	At capacity	WAITLISTED
6	= 0	No	All met	Fits	Below capacity	REGISTERED

Priority interaction cases include dues + duplicate, duplicate + missing prerequisites, missing prerequisites + credit limit, and credit limit + full course. These verify precedence rather than only isolated outcomes.

Important observed behavior: the current code blocks dues only when the balance is greater than Rs.1000. The SRS decision table specifies any amount above zero. Tests retain the Rs.1 expected DUES_BLOCK result.

3. CFG: registration_decision(student, course)

Actual implementation flow; red labels show early returns/exceptions. Dashed waitlist edge is infeasible after validation.



Validation enforces enrolled <= capacity, so D7 true cannot occur.

4. Complexity, basis paths, and MC/DC

Cyclomatic complexity under the assignment's seven-if-decision convention is M = 1 + 7 = 8. The compound capacity/enrollment expression is counted as one if decision. This is a structural count; the waitlist basis path is infeasible.

Path	Decision sequence / outcome
P1	D1 true -> Invalid dues exception
P2	D1 false, D2 true -> Invalid capacity/enrollment exception
P3	D1-D2 false, D3 true -> DUES_BLOCK
P4	D1-D3 false, D4 true -> ALREADY_REGISTERED
P5	D1-D4 false, D5 true -> PREREQUISITES_MISSING
P6	D1-D5 false, D6 true -> CREDIT_LIMIT
P7	D1-D6 false, D7 true -> WAITLISTED (structural, infeasible)
P8	All decisions false -> REGISTERED

Compound condition analysis

Let A = enrollment outside 0..capacity; B = capacity < 1; decision = A OR B. WB03 (capacity 2, enrolled 1) gives (F,F) -> F. WB02A (capacity 2, enrolled -1) gives (T,F) -> T. WB02C (capacity 0, enrolled 0) gives (F,T) -> T. WB02D has logical (T,T), but Python short-circuits and does not evaluate B when A is true.

MC/DC is demonstrated for this compound decision: compare WB03 with WB02A to show A changes the outcome while B is false; compare WB03 with WB02C to show B changes the outcome while A is false. Each pair changes only one condition. This claim is limited to this compound condition.

5. Coverage, defects, and repository collaboration

Coverage scope	Statements	Branches	Reported result
Combined registration.py	51/53 (96.2%)	24/26 arcs (92.3%), 2 partial	95% combined pytest-cov
registration_decision()	14/15 (93.3%)	13/14 (92.9%)	WAITLISTED line/true edge uncovered

Confirmed defects in the original implementation

Observed defect	SRS rule	Status
Exactly 18 credits rejected	FR-01: total <= 18 allowed	Unresolved; boundary test fails
Partial prerequisite match accepted	FR-03: all required courses completed	Unresolved; ECP test fails
Dues blocked only above Rs.1000	FR-04: any positive dues blocks	Unresolved; Rs.1 tests fail
Full course returns REGISTERED	FR-08: full course -> WAITLISTED	Unresolved; waitlist path unreachable
Withdrawal leaves enrollment unchanged	FR-13: decrement enrollment	Unresolved; state test fails

GitHub collaboration / PR evidence

A GitHub origin and branch refs (main, blackbox-testing, whitebox-testing) are present in local repository metadata. Inspected repository files contain no PR URL, issue record, review/approval, or named contribution log. Thus PR creation, review status, issue identifiers, and member contributions cannot be verified from available repository evidence. README guidance describing an intended workflow is not evidence that it occurred.

Evidence: initial, expanded, white-box, and combined test transcripts plus coverage XML are retained under docs/evidence/. CI workflow is configured to run pytest with branch coverage on pushes and pull requests; genuine expected failures remain visible.